	Academic Semester	Fall 2026-2027	**Academic level**	1
	Course Code	MA104	**Exam**	Quiz 1 — Model F
	Course Title	Discrete Mathematics		
	Time Allowed	30 Minutes	**Total Points**	10
	Course Instructors	Dr. Salah Moawad , Dr. Tamer Seuodi		

Student Name: _____ **Student ID:** _____ **Section:** _____

*Answer **all** questions. Use T / F in truth tables and 1 / 0 in membership tables.*

Question 1 — Choose the correct answer and give the reason (5 points)

One point each: circle the letter of the correct answer **and** write a short reason in the box.

1. If p is **F** and q is **T**, which proposition is **false**?

A) $p \vee q$ **B)** $p \to q$
C) $p \leftrightarrow q$ **D)** $\neg p \wedge q$

Reason:

2. Let $A = \{2, 3\}$ and $B = \{3, 6, 9\}$. Then $A \cup B =$

A) $\{3\}$ **B)** $\{2, 6, 9\}$
C) $\{2, 3, 6, 9\}$ **D)** $\{2\}$

Reason:

3. $p \to q \equiv$?

A) $\neg q \to \neg p$ **B)** $q \to p$
C) $\neg p \to \neg q$ **D)** $p \wedge \neg q$

Reason:

4. Is $p(q \wedge r)$ a **well-formed formula** (WFF)? If not, name the rule it breaks.

A) WFF **B)** Not a WFF

Reason:

5. Let $U = \{1, 2, ..., 7\}$, $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5\}$. Then $A \cap B^c =$

A) $\{3, 4\}$ **B)** $\{5\}$
C) $\{1, 2\}$ **D)** $\{1, 2, 6, 7\}$

Reason:

Question 2 — Truth table (2.5 points)

(a) Complete the truth table of $((p \leftrightarrow q) \wedge \neg p) \to q$. [2 pts]

p	q	$p \leftrightarrow q$	$\neg p$	$(p \leftrightarrow q) \wedge \neg p$	$((p \leftrightarrow q) \wedge \neg p) \to q$
T	T				
T	F				
F	T				
F	F				

(b) Let P be the proposition in (a). Is $P \equiv p \vee q$? Justify using your table. [½ pt]

Question 3 — Membership table (2.5 points)

(a) Complete the membership table for the identity $(A \cup B)^c = A^c \cup B^c$. [2 pts]

A	B	$A \cup B$	$(A \cup B)^c$	A^c	B^c	$A^c \cup B^c$
1	1					
1	0					
0	1					
0	0					

(b) Is the identity $(A \cup B)^c = A^c \cup B^c$ **true** or **false**? Justify using your table. [½ pt]

— End of quiz · Good luck —